[MMS Test Study] AI Department Session 6: Core Architectures & Limitations

1. Convolutional Neural Networks (CNN) - Spatial Representation

CNN architectures are designed to extract spatial features directly from grid-structured inputs like images. Unlike traditional fully connected layers (Dense Layers) which flatten structural data, CNN preserves spatial correlation using Local Receptive Fields. The key operations consist of: (1) Convolutional Filter: sliding learnable kernels to generate feature maps, reducing parameters via Shared Weights. (2) Pooling Layer: Downsampling feature maps (e.g., Max Pooling) to provide translation invariance and decrease computational complexity. However, CNNs have limitations in capturing temporal sequence variations and require significant spatial context tuning.

2. Recurrent Neural Networks (RNN) - Sequential Processing

RNN is optimized for sequential data such as natural language and time-series forecasting. It introduces a feedback loop where the hidden state at time 't' depends on both the input at time 't' and the hidden state from 't-1'. This allows the model to retain a memory of previous steps. The critical limitation of standard RNN is the Vanishing and Exploding Gradient problems during Backpropagation Through Time (BPTT). When dealing with long sequences, gradients propagated back in time either shrink to zero or grow exponentially, making it impossible to capture long-term dependencies.

3. Long Short-Term Memory (LSTM) - Gated Architecture

LSTM addresses the vanishing gradient problem of vanilla RNNs by introducing a specialized gated mechanism and a Cell State (C_t). The Cell State acts as a conveyor belt, allowing information to flow through with minimal modifications. The flow is regulated by three distinct gates: (1) Forget Gate: decides what information to discard from the previous cell state. (2) Input Gate: determines which new information to store in the current cell state. (3) Output Gate: controls what information from the updated cell state to output as the hidden state. While highly effective for moderate sequences, LSTM struggles with extreme sequence lengths and is computationally expensive due to its sequential, non-parallelizable nature.